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Predicting Income Levels Using Machine Learning and Explainable AI: Uncovering the Drivers of Economic Opportunity

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I. INTRODUCTION

INCOME is a major indicator of economic well-being and socioeconomic opportunity. In particular, it determines whether individuals and households can meet basic needs and access essential services. Moreover, income differences reflect unequal access to education, employment, technology, and other resources. In this regard, previous studies have used personal and socioeconomic characteristics to estimate income and examine income-related outcomes [1]–[3]. Therefore, identifying the factors associated with income levels is important for understanding socioeconomic inequality.

Digital access also affects participation in economic activities. More specifically, internet access and digital technologies connect individuals to information, employment opportunities, training, and online economic activities.. For example, research in Indonesia has shown that digital access and internet use are associated with income and productivity [4]. Similarly, research in Brazil has used machine learning and SHAP values to examine the relationship between ICT use and income distribution [5]. These findings suggest that digital access can improve opportunities to obtain information, develop skills, and participate in higher-value economic activities.

Demographic and socioeconomic characteristics also contribute to income differences. Education can improve access to employment opportunities, while occupation and employment conditions can affect earning potential. Furthermore, regional characteristics may reflect differences in labor markets, infrastructure, and access to economic resources. Accordingly, previous studies have used personal attributes, occupations, and socioeconomic characteristics for income prediction and analysis [3], [6], [7]. Similarly, labor-market studies in Indonesia have identified differences in employment outcomes across population groups [8]. Taken together, these findings support the use of demographic, educational, employment, and regional characteristics as complementary income predictors.

Machine learning can model complex relationships among these characteristics. Unlike simple models that often assume linear relationships, machine learning algorithms can identify interactions and nonlinear patterns across multiple features.

Several studies have applied machine learning to income, poverty, and income-inequality prediction [1], [9], [10]. For example, Ji used machine learning to predict income inequality, whereas Nkurunziza *et al.* compared multiple classifiers for poverty classification [9], [10]. These studies show that machine learning can combine diverse socioeconomic indicators to distinguish individuals or households with different economic outcomes.

However, income classification remains insufficiently studied in developing countries. Existing research has often focused on binary income thresholds, poverty prediction, or specific population groups [6], [9], [11]. Although these approaches provide useful information, they may not capture differences among several income groups. In many developing countries, lower- and middle-income groups contain most of the population, whereas the upper-income group contains far fewer observations. Consequently, multiclass prediction is difficult because the model must distinguish several groups while also identifying a relatively small upper-income class.

Class imbalance can further reduce the reliability of income-class predictions. When one class contains more observations than another, a model may favor the majority class during training. As a result, a model can achieve high overall accuracy but produce weak predictions for minority income groups. Previous research has shown that imbalanced data can reduce the effectiveness of standard learning algorithms [12]. SMOTE addresses this problem by generating synthetic observations for minority classes [13]. In contrast, ADASYN focuses more strongly on difficult minority observations [14]. Therefore, evaluating these strategies can improve predictive performance across income classes.

The choice of algorithm and model configuration also affects income-classification performance. Different algorithms represent socioeconomic relationships in different ways. Random Forest combines multiple decision trees to produce robust predictions [15]. XGBoost uses a scalable tree-boosting framework for structured-data prediction [16]. Logistic Regression provides a simpler classification model and can serve as a baseline [17]. Moreover, hyperparameter optimization can improve model performance by identifying suitable configurations for the data [18]. Therefore, comparing and optimizing multiple algorithms is necessary to determine suitable approaches for income-class prediction.

Predictive performance alone does not explain the socioeconomic factors behind model decisions. This limitation is important because government agencies, policymakers, and development organizations need to understand why individuals are assigned to particular income classes. Explainable artificial intelligence provides methods for interpreting predictions and identifying influential features [19]–[21]. In addition, previous research has applied explainable machine learning to public policy and socioeconomic analysis [22], [23]. Therefore, interpretable models can connect predictions with education, employment, digital access, and other characteristics relevant to socioeconomic interventions.

Recent studies have demonstrated the usefulness of explainable AI for income-related analysis. Bai *et al.* used SHAP to analyze factors associated with income estimation in New York City [24]. Carrillo *et al.* used individual explanations to examine poverty estimation and interpret predictions at the individual level [25]. Herrera *et al.* examined ICT use and income distribution using SHAP values [5]. Jahan *et al.* combined LightGBM with explainable AI for adult income prediction [26]. Similarly, Komara *et al.* applied machine learning and explainable AI to income-poverty risk using recent European survey data [11]. These studies show that explainable AI can identify features associated with predictions instead of reporting predictive performance alone.

Nevertheless, a research gap remains in multiclass income prediction for developing countries. Existing studies have mainly examined binary income prediction, poverty estimation, income inequality, or specific population groups [6], [9]–[11]. Relatively little research has examined multiple income classes using recent national survey data while jointly considering digital access, class imbalance, and explainability. This gap is especially important when lower- and middle-income groups represent most observations and the upper-income group represents a much smaller proportion. Therefore, a suitable framework must address class imbalance and provide interpretable socioeconomic insights.

This study addresses this gap by applying machine learning to an Indonesian survey dataset derived from the August 2023 National Labor Force Survey (SAKERNAS) [27]. The study predicts four income classes using demographic, educational, employment, regional, and digital-access characteristics. First, multiple machine learning algorithms are trained and compared through systematic hyperparameter optimization. Next, SMOTE and ADASYN are evaluated to address class imbalance. Then, the strongest individual models are combined using hard- and soft-voting ensembles. Finally, SHAP-based explanations are used to identify the features that contribute most to income-class predictions. The workflow includes data preprocessing, categorical-feature encoding, feature scaling where required, stratified data partitioning, model training, hyperparameter tuning, imbalance mitigation, class-sensitive evaluation, ensemble construction, and post hoc interpretation.

The primary goal of this study is to determine whether the digital index is the most influential factor in distinguishing individuals across income classes. Specifically, the study examines whether the digital index has a greater contribution to income-class predictions than demographic, educational, employment, and regional characteristics. This goal is evaluated through model-based feature-importance analysis and SHAP explanations, which quantify the overall and class-specific influence of the digital index on the predictions. Accordingly, the central research hypothesis is that the digital index has the highest predictive impact on income-class assignment. The analysis also allows this hypothesis to be assessed objectively if other socioeconomic characteristics demonstrate greater influence.

The proposed framework can support targeted socioeconomic analysis. For example, government agencies may use the identified patterns to recognize groups that require economic, employment, or digital interventions. Policymakers can also examine whether education, occupation, employment conditions, regional characteristics, or digital access are associated with particular income classes. In addition, development organizations may use these findings to design programs related to digital literacy, technical training, and employment support. These applications do not replace policy evaluation. However, they can provide evidence for directing resources toward groups with greater socioeconomic needs.

This study develops an integrated framework for multiclass income prediction and interpretation. The framework combines model comparison, hyperparameter optimization, class-imbalance handling, ensemble learning, and explainable AI. Its objective is to produce reliable predictions, determine whether the digital index is the most influential predictor, and identify the characteristics associated with different income classes. By applying the framework to recent Indonesian survey data, this study provides a data-driven perspective on income disparity and the potential role of digital and socioeconomic characteristics in shaping economic opportunity.

The main contributions of this study are summarized below:

- Application of machine learning to a relatively under-explored Indonesian survey dataset derived from the August 2023 National Labor Force Survey (SAKERNAS) for multiclass income prediction and socioeconomic analysis.
- Development of a complete machine learning framework for predicting four income classes using demographic, educational, employment, regional, and digital-access features.
- Systematic comparison and hyperparameter optimization of Logistic Regression, Random Forest, XGBoost, CatBoost, and LightGBM for income-class prediction.
- Evaluation of SMOTE and ADASYN for improving the prediction of minority income classes in an imbalanced income distribution.
- Development and evaluation of hard-voting and soft-voting ensemble models using the strongest individual

classifiers.
- Investigation of whether the digital index has the greatest impact on income-class prediction compared with demographic, educational, employment, and regional characteristics.
- Application of explainable AI to identify influential features, quantify the contribution of the digital index, and provide interpretable socioeconomic insights.
- Analysis of digital-access and broader socioeconomic characteristics to support targeted skill development, employment programs, and socioeconomic interventions.

Together, these contributions establish a combined predictive and explanatory approach to income-class analysis. The resulting insights can improve understanding of socioeconomic disparity, empirically assess the role of digital access in income-class assignment, and inform targeted efforts to expand economic opportunities in developing-country contexts.

The remainder of this report is organized as follows: Section II describes the dataset, data pre-processing, exploratory data analysis, data imbalance handling techniques, and the ensemble modeling approach. The section III represents experimental findings, including confusion matrices, a precision-recall curve, an ROC curve, and explainable AI analysis. Finally, section IV discusses key findings, including subgroup analysis, interaction analysis, disability insight, upper-income boundary sensitivity analysis, and conclusion.

II. METHODOLOGY

A. DATASET

The dataset used in this project was obtained from the National Labor Force Survey [27], conducted by BPS-Statistics Indonesia and released in August 2023. Moreover, the dataset contains harmonized micro-level labor force data covering agricultural employment, disability status, digital technology use, social participation, income, working hours, education, and other socio-demographic information. These variables are important for this study because they enable the examination of the relationship between digital technology use and income class while considering other relevant factors. In addition, the dataset includes agricultural employment and productivity-related information. This is useful because income class can vary between agricultural and non-agricultural workers. Therefore, the dataset provides a suitable basis for examining whether digital technology use is associated with differences in income class.

B. DATA PRE-PROCESSING

To ensure the quality, consistency, and reliability of the dataset, a systematic data pre-processing procedure was undertaken. The preprocessing procedure involved several sequential steps to identify and address data quality issues prior to model development.

1) Handling missing values

The raw dataset contained substantial missingness across several features. The highest level of missingness was observed for disability_type (92.20%), followed by internet_use (66.80%), while weekly_workhours and monthly_workhours had the lowest proportion of missing values (33.16%) as illustrated in as shown in Table 1. Rather than uniformly dropping high-missingness columns, this study treated missingness as informative. For example, for weekl_workhours, missing values were set to 0 and flagged with a binary indicator (workhours_missing), as these records correspond to individuals who have a job but were temporarily not working during the reference week, a category explicitly recognized in SAKERNAS [28]. Similarly, missing monthly_income was not dropped but mapped to a dedicated No_income category during target construction to preserve the distinction between "no income" and "unreported income." Moreover, SAKERNAS separately identifies individuals who are temporarily not working and records whether they continue to receive income or wages during that period, indicating that an absence of reported income can have substantive labor-market meaning rather than necessarily representing a data-quality issue. Redundant income-related fields (monthly_income, log_monthly_income, agri_productivity, agri_productivity_trimmed, monthly_workhours) were dropped once their information was captured by derived features to avoid the need for model-based imputation on economically sensitive variables.

TABLE 1. Missingness of Selected Variables

Variable	Missing (%)
disability_type	92.20
internet_use	66.80
weekly_workhours	33.16
monthly_workhours	33.16

2) Categorical and binary encoding

We encoded the categorical variables according to their measurement type to obtain numerical representations while preserving the information and structure of the original variables. For nominal variables with two levels, we used binary indicators, as summarized in Table 2. This approach represents the presence of a category using a value of 1 and its absence using 0.

For the ordinal variable *disability_severity*, we assigned integer values according to the increasing severity of disability: non-disabled = 0, mild = 1, severe = 2, and total = 3. Thus, the original ordering was retained in the numerical representation rather than treating the categories as unordered. We also encoded *disability_type* using separate binary indicator variables for vision, mental/other, walking, hearing, hand/finger, and communication difficulties. In addition, we created a *no_disability* indicator for observations with no recorded disability type. This representation preserves the specific disability characteristics of each observation and facilitates their interpretation in the subsequent machine learning analysis.

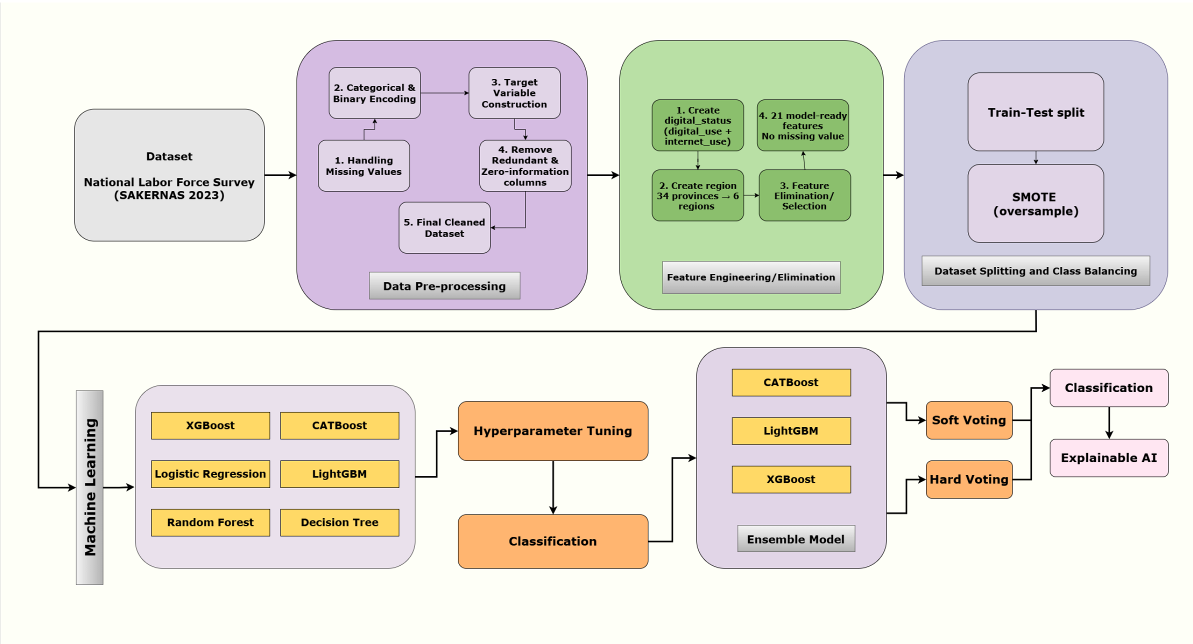


FIGURE 1. Schematic representation of the end-to-end methodology for income-level prediction using SAKERNAS 2023 data. The workflow includes data preprocessing, feature engineering and selection, train–test splitting, SMOTE-based class balancing, machine learning model development and hyperparameter tuning, ensemble construction using soft and hard voting, and final income-level classification with explainable artificial intelligence (XAI).

TABLE 2. Binary Encoding of Selected Categorical Variables

Original Variable	Encoded Variable
Gender	gender_male
Disability	disabled
Digital use	is_digital_user
Social participation	social_participant
Agricultural employment	agri_worker

3) Feature engineering

Two composite categorical features were engineered from combinations of raw variables. First, *digital_status* was derived from *digital_use* and *internet_use* to produce a three-level ordinal variable (0 = non-digital user, 1 = digital user without internet access, 2 = internet user), combining two related but separately coded variables into a single interpretable measure of digital inclusion. Second, the 34-level province variable was coarsened into a six-level region variable (Sumatra, Java, Bali_Nusa, Kalimantan, Sulawesi, and Eastern) based on the official province codes provided by BPS-Statistics Indonesia [29], as shown in Table 3. This transformation reduced dimensionality while retaining meaningful geographic variation.

4) Target variable construction

The prediction target, income_class, was constructed from monthly_income using expenditure-based income thresholds adapted from BPS 2024 household classification bands (Middle class lower bound: IDR 2,040,262; Upper class lower bound: IDR 9,909,844) [30], which yielded four classes: No_income, Lower, Middle, and Upper, as shown in Table 4.

TABLE 3. Mapping of Indonesian Province Codes to Geographical Regions

Code	Province (Region)
11–19, 21	Sumatra (Sumatra)
31–36	Java (Java)
51–53	Bali & Nusa Tenggara (Bali_Nusa)
61–65	Kalimantan (Kalimantan)
71–76	Sulawesi (Sulawesi)
81–82	Maluku (Eastern)
91, 92, 94–97	Papua (Eastern)

Records with missing or non-positive income were assigned to the No_income class rather than excluded, since this reflects a substantively meaningful labor-market status (e.g., non-participation) rather than a data-quality issue.

TABLE 4. Income Class Definition

Income Class	Monthly Income (IDR)
No_income	≤ 0 or missing
Lower	> 0 and $< 2{,}040{,}262$
Middle	2,040,262–9,909,843
Upper	$\geq 9{,}909{,}844$

5) Redundant and zero-information columns

Once the engineered features were created, their original source columns were removed to avoid redundant information. These included the categorical source variables *gender*, *disability*, *disability_severity*, *disability_type*, *digital_use*, *internet_use*, *social_participation*, *agri_employment*, and *province*. We also removed redundant numeric variables, including *agri_productivity*, *agri_productivity_trimmed*, *log_monthly_income*, *monthly_workhours*, and *monthly_income*, as well as the temporary helper variable *is_digital_user*. After these preprocessing steps, the final dataset contained 21 model-ready features with no missing values across the retained columns.

6) Final dataset

The cleaned dataset comprises 751,622 records and 21 features spanning demographic (*age*, *education_years*, *gender_male*), disability (*disabled*, *disability_severity_ord*, disability-type flags), labor and digital inclusion (*weekly_workhours*, *workhours_missing*, *digital_status*, *digital_index*), social participation (*social_participant*, *social_participation_index*), agricultural employment (*agri_worker*), geographic (*region*), and the target (*income_class*) variables.

C. EXPLORATORY DATA ANALYSIS

The final preprocessed dataset used for this study consists of 751,622 observations across 26 variables. The target variable is income_class, which is a categorical variable that has four distinct classes: No_income, lower, middle, and upper. However, the dataset contains significant class imbalance with No_income comprises 43.63% (n = 327,963), Lower for 31.85% (n = 239,425) of observations, Middle for 23.40% (n = 175,908) and Upper only 1.11% (n = 8,326). The target distribution is shown in Figure 2. Target distribution The 26

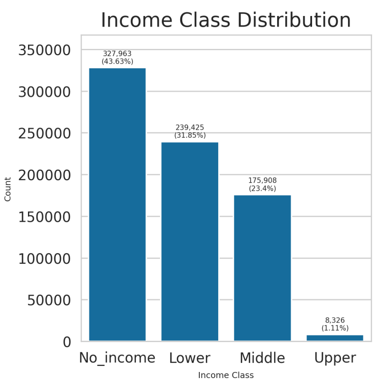


FIGURE 2. Income class distribution.

predictor variables span several categories:

- **Demographic:** age, gender_male
- **Socioeconomic/labor:** education_years, weekly_workhours, agri_worker, workhours_missing
- **Digital access:** digital_index, digital_status
- **Social participation:** social_participation_index, social_participant
- **Disability status:** disabled, disability_severity_ord, no_disability, and six disability-type indicators (vision, mental, walking, hearing, communication, hand/finger)
- **Regional:** six one-hot indicators for region (Bali_Nusa, Eastern, Java, Kalimantan, Sulawesi, Sumatra)

Additionally, there are no missing values in this dataset. The upper-income group has a higher median number of years of education, indicating a positive association between educational attainment and upper-income class membership. Similarly, the median values of the digital index and social participation index are substantially higher for the upper-income group. The upper-income group also shows a moderately higher median age and weekly working hours, although the distributions exhibit considerable overlap with the other group. The continuous features show distinct curves. Age is right-skewed, with most individuals concentrated between 20 and 50 years, followed by a gradual decline in frequency among older age groups, particularly those aged 90 years and above. Education_years shows clear spikes at specific values, especially 1, 6, 9, and 12 years. This suggests that the variable reflects distinct levels of formal education rather than a continuous measure of schooling. Weekly_workhours has a very large concentration at zero, which is consistent with individuals who are not working or have no income. Among those who work, the distribution extends to more than 90 hours per week. Digital_index, digital_status, and disability_severity_ord range are low-cardinality ordinal variables as they range from 0 to 3. Each variable is strongly concentrated at zero or low values with progressively fewer observations at higher levels. Finally, social_participation_index is mainly concentrated around 0.4, with smaller groups near 0 and 0.65. This pattern suggests that the index is based on a limited number of discrete participation categories rather than representing a fully continuous measure. Among the binary variables, the prevalence varies considerably. No_disability is highly common, with 92.0% of individuals classified as having no disability, while social_participant is also dominant at 83.0%. Gender_male is balanced with 49.0% of the sample. In contrast, the individual disability-type indicators, including vision, mental, walking, hearing, communication, and hand/finger disabilities, are relatively rare, each having less than 3% of the sample. Due to their low prevalence, these individual disability indicators may provide limited predictive power when considered in isolation. Therefore, combining them or using broader measures such as disabled and disability_severity_ord may provide a more meaningful representation of disability status. Regionally, the sample is dominated by Sumatra and Java, followed by Sulawesi, Kalimantan, Eastern, and Bali_Nusa in descending order. A Pearson correlation analysis of the continuous features revealed no severe multicollinearity overall. Moderate positive correlations were observed between education_years and digital_index (r = 0.43), weekly_workhours and digital_status (r = 0.48), and social_participation_index and digital_index (r = 0.38). This suggests that education, employment intensity, and social participation are all loosely linked to digital access. Age showed a mild negative correlation with education_years (r = -0.33) which is consistent with generational differences in educational attainment. The correlation structure is shown in Fig. 3. Furthermore, digital_index shows a clear and consistent relationship with income_class as mean digital_index rises monotonically across the ordered classes, from 0.08 in

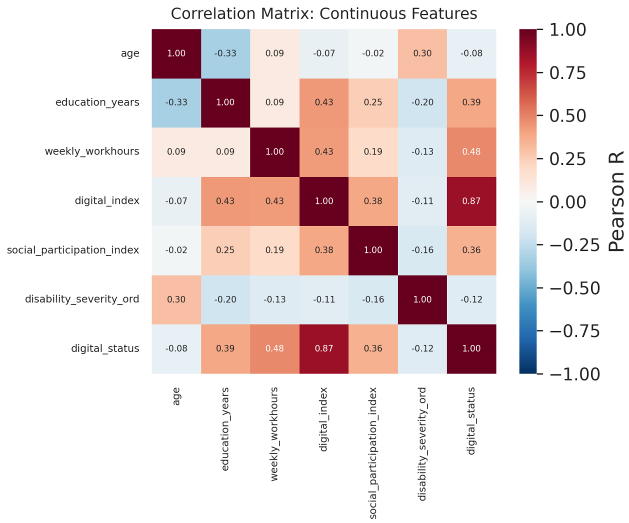


FIGURE 3. **Pearson correlation matrix of continuous features.**

No_income, to 0.55 in Lower, 1.12 in Middle, and 1.80 in Upper, with non-overlapping 95% confidence intervals at each step. This visual pattern is shown in Fig. 4.

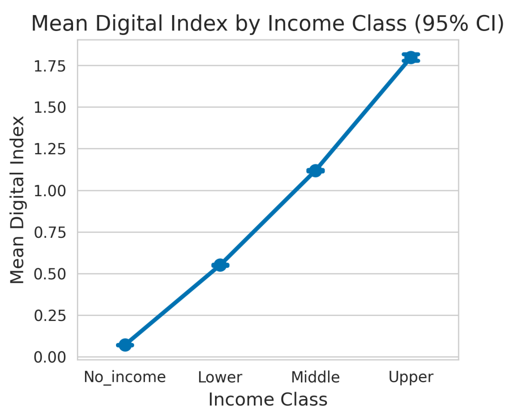


FIGURE 4. **Mean digital index by income class with 95% confidence intervals.**

Moreover, this pattern aligns with the multinomial logistic regression (MNLogit) result established earlier in the project which reinforces digital_index as one of the strongest and most reliable predictors of income class in this dataset. The distribution of digital_index by income class is shown in Figure 6.

D. MACHINE LEARNING ALGORITHMS

1) Logistic Regression

Logistic Regression is a supervised classification algorithm that can be applied to both binary and multiclass classification problems.It models the probability of a binary outcome using the logistic function [17].

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)}} \tag{1}$$

in (1), β_0 represents the intercept, β_i denotes the coefficient for the i-th feature, and p indicates the number of features. The model estimates its parameters using maximum likelihood estimation, making it interpretable through coefficient analysis. In this experiment, the machine learning model created with this machine learning algorithm produced the confusion matrix in Fig. 7d.

2) Decision Tree

A Decision Tree is a supervised learning algorithm which is used to solve classification problems.The confusion matrix in Fig. 7b was obtained using Decision Tree.

3) Random Forest

Random Forest constructs multiple decision trees using bootstrap sampling and random feature selection [15]. The predictions of the individual trees are aggregated through majority voting:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}) \tag{2}$$

in (2), B represents the number of trees, $T_b(\mathbf{x})$ denotes the prediction from the b-th tree, and $\mathbf{x}$ represents the input features. Furthermore, feature importance is calculated using the reduction in Gini impurity:

$$\text{Importance}(X_j) = \sum_{t \in T} p(t) \Delta i(t, X_j) \tag{3}$$

in (3), $p(t)$ is the proportion of samples reaching node t, and $\Delta i(t, X_j)$ quantifies the reduction in impurity achieved by splitting on feature X_j.

The confusion matrix obtained from Random Forest is shown in Fig. 7e.

4) XGBoost

The eXtreme Gradient Boosting (XGBoost) is a high-performance gradient boosting machine learning algorithm used mainly for classification and regression tasks [31]. By combining multiple decision trees as shown in (4), XGBoost can achieve high predictive accuracy.

$$\hat{y} = \sum_{k=1}^{K} f_k(x) \tag{4}$$

In this experiment, the machine learning model created with this machine learning algorithm produced the confusion matrix in Fig. 7h.

5) LightGBM

Light Gradient Boosting (LightGBM) is a gradient boosting algorithm for regression and classification applications, which is fast and memory-efficient. Also, it creates an ensemble of decision trees . Moreover, LightGBM is optimized for large-scale datasets, emphasizing performance and memory efficiency. The confusion matrix in Fig. 7c was obtained using LightGBM.

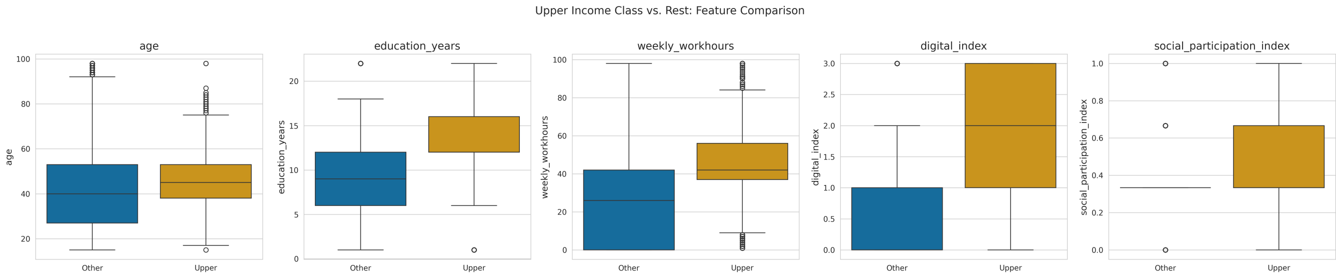


FIGURE 5. **Upper income class versus the remaining observations: feature comparison.**

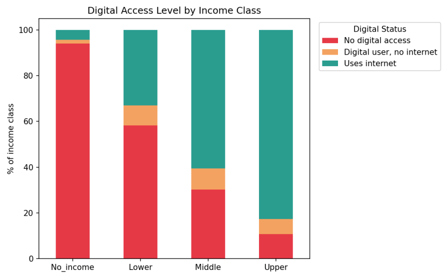


FIGURE 6. **Digital index by income class.**

6) CatBoost

Categorical Boosting (CatBoost) is a gradient boosting machine learning algorithm for supervised learning tasks such as classification and regression. Although, it is similar to XGBoost and LightGBM, it is particularly known for handling categorical features effectively without requiring extensive preprocessing. In this experiment, the machine learning model created with this machine learning algorithm produced the confusion matrix in Fig. 7a.

E. HYPERPARAMETER TUNING

Hyperparameter tuning is an essential step in developing effective machine learning models. Basically, It involves finding a suitable set of hyperparameters to improve model performance. Since the number of possible hyperparameter combinations can be large, RandomizedSearchCV was used instead of GridSearchCV to efficiently explore the hyperparameter space by evaluating a fixed number of randomly selected parameter combinations [18], [32]. Table 5 presents the models, hyperparameter spaces explored, and the selected hyperparameter values.

F. ENSEMBLE MACHINE LEARNING MODEL CONSTRUCTION

Ensemble learning involves combining multiple machine learning models to improve predictive performance, generilzation, and reduce overfitting, [33]. In this study, the three best-performing models based on macro F1-score, as reported in Table 6, namely CatBoost Classifier, XGBoost Classifier, and LightGBM Classifier, were selected to construct the ensemble. Two ensemble methods were considered: hard voting and soft voting [34]. Hard voting, also known as majority voting, selects the class that receives the most predictions from the individual models. Mathematically, for an ensemble of n models, the hard voting prediction $\hat{y}$ for an instance x can be represented as:

$$\hat{y} = \mathrm{mode}(\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n) \tag{5}$$

where $\hat{y}_i$ represents the prediction of the i-th model.

Soft voting, in contrast, uses the predicted class probabilities from the individual models. The probabilities are averaged across the models, and the class with the highest average probability is selected as the final prediction. The soft voting prediction $\hat{y}$ for an instance x is given by:

$$\hat{y} = \arg\max_{c} \left(\frac{1}{n} \sum_{i=1}^{n} P_i(y = c \mid x) \right) \tag{6}$$

where $P_i(y = c \mid x)$ denotes the predicted probability of class c from the i-th model.

G. EXPERIMENTAL DESIGN

1) Train-Test Split

The dataset was split into training and test sets using an 80:20 ratio. Accordingly, 80% of the observations were used as the training set for model training, while the remaining 20% were reserved as the test set for final model evaluation. The dataset was shuffled before splitting to reduce the potential influence of the original data ordering on the resulting subsets. Also, a fixed random seed was used to ensure full reproducibility of the partition.

2) Class Imbalance Handling

The target variable has a substantial class imbalance, with the upper-income class having far fewer observations than the other classes, as illustrated in Fig. 2. Thus, class imbalance was considered during model training to reduce bias toward the majority classes. Different class-balancing strategies were evaluated using the training data, and the best-performing strategy was selected for the final experiments. The detailed comparison of these strategies is presented in Section III-B.

3) Model Training and Evaluation

The models were trained on the training data after preprocessing and class-imbalance handling. Logistic Regression, Decision Tree, Random Forest, XGBoost, CatBoost, and

TABLE 5. Hyperparameter Search Spaces and Selected Values

Model	Hyperparameter	Hyperparameter Space	Selected
DecisionTreeClassifier	max_depth	{5, 10, 15, 20}	20
	min_samples_split	{2, 5, 10}	2
	min_samples_leaf	{1, 2, 4}	2
RandomForestClassifier	n_estimators	{100, 150, 200}	100
	max_depth	{10}	10
	min_samples_split	{2, 5, 10}	10
	min_samples_leaf	{1, 2, 4}	4
XGBClassifier	n_estimators	{100, 150, 200}	200
	max_depth	{6, 8, 10}	10
	learning_rate	{0.01, 0.05, 0.1}	0.05
LogisticRegression	C	{0.01, 0.1, 1.0, 10.0}	10.0
	max_iter	{1000}	1000
CatBoostClassifier	iterations	{200, 400, 600}	400
	depth	{6, 8, 10}	10
	learning_rate	{0.01, 0.05, 0.1}	0.05
	l2_leaf_reg	{3, 5, 7}	7
LGBMClassifier	n_estimators	{100, 200, 300}	300
	max_depth	{6, 8, 10}	8
	learning_rate	{0.01, 0.05, 0.1}	0.1
	num_leaves	{31, 50, 70}	50

LightGBM were trained and evaluated using the same experimental setup to ensure a fair comparison. Hyperparameters were optimized using RandomizedSearchCV with five-fold stratified cross-validation and macro F1-score as the selection metric. The final models were then evaluated on the unseen test data.

Model performance was assessed using accuracy, macro F1-score, and ROC AUC. Accuracy measures the overall proportion of correctly classified observations, whereas macro F1-score gives equal importance to all income classes. ROC AUC measures the model's ability to distinguish between the income classes. These metrics were used together to provide a balanced evaluation, particularly because the target variable is imbalanced.

III. EXPERIMENTAL RESULTS

A. OVERALL MACHINE LEARNING ALGORITHMS PERFORMANCE

Table 6 illustrates the performance metrics (accuracy, macro-averaged F1-score, precision, recall, and ROC-AUC) for six machine learning models, and two ensemble models. Moreover, we performed 5-fold cross-validation on eight ML models during hyperparameter tuning and selected the hyperparameter configuration that achieved the highest macro F1-score for each model [35].

The performance comparison in Table 6 shows that the evaluated models achieved relatively similar overall performance, with clear differences across individual evaluation metrics. Among the models, the LGBM Classifier achieved the highest accuracy of 74.97% and the highest precision of 60.85%. However, CatBoost achieved the highest F1-score of 59.90%, indicating the best balance between precision and recall among the evaluated models. Therefore, no single standalone model dominated across all performance measures.

The ensemble models also produced competitive results. In particular, the Soft Voting ensemble achieved an accuracy of 74.83% and an F1-score of 59.84%, which were close to the best standalone results. More importantly, it achieved the highest ROC-AUC of 90.87%, indicating strong overall discrimination between the income classes. The Hard Voting ensemble also performed competitively, achieving 74.78% accuracy and 59.75% F1-score. However, its ROC-AUC could not be reported because hard voting produces discrete class predictions rather than class [36], [37]. Thus, the Soft Voting ensemble provides a more informative probabilistic assessment of class discrimination while maintaining competitive classification performance.

Among the individual models, XGB achieved an accuracy of 74.61%, an F1-score of 59.68%, and a ROC-AUC of 90.67%. Similarly, CatBoost and LGBM achieved ROC-AUC values of 90.81% and 90.72%, respectively. These results indicate that the boosting-based models were particularly effective at distinguishing between the income classes. Their strong ROC-AUC values also suggest that the models can rank observations across classes effectively, even when their F1-scores remain comparatively modest.

In contrast, the Decision Tree Classifier produced a lower F1-score of 57.49% and a ROC-AUC of 84.30%. Logistic Regression achieved a similar F1-score of 56.40%, although its recall reached 61.61%. Random Forest produced the highest recall of 62.40%, but its accuracy and F1-score were lower than those of the boosting-based models. These results show that a higher recall does not necessarily translate into better overall classification performance, particularly when the objective is to maintain a balance between precision and recall.

Overall, the results in Table 6 indicate that the boosting-based models and the Soft Voting ensemble provided the most competitive performance across the evaluated metrics. LGBM achieved the highest accuracy and precision, CatBoost achieved the highest F1-score, while the Soft Voting ensemble

TABLE 6. Performance comparison of the evaluated machine learning models.The best results for each performance metric is highlighted in bold.

Model	Accuracy (%)	F1-Score (%)	Precision (%)	Recall (%)	ROC-AUC (%)
Decision Tree Classifier	72.70	57.49	57.65	57.70	84.30
Random Forest Classifier	70.95	56.74	56.85	**62.40**	89.45
XGB Classifier	74.61	59.68	60.21	59.65	90.67
Logistic Regression	70.59	56.40	56.07	61.61	88.85
CatBoost Classifier	74.58	**59.90**	60.15	60.11	90.81
LGBM Classifier	**74.97**	59.46	**60.85**	58.88	90.72
Ensemble Model (Hard Voting)	74.78	59.75	60.46	59.56	–
Ensemble Model (Soft Voting)	74.83	59.84	60.66	59.58	**90.87**

achieved the highest ROC-AUC. This variation across metrics highlights the importance of considering multiple evaluation measures rather than relying on accuracy alone, particularly for a multiclass income prediction problem.

B. RESAMPLING STRATEGY COMPARISON

To address the substantial class imbalance in the target variable, we compared five class-balancing strategies: class weighting, SMOTE [13], SMOTE with Tomek Links, ADASYN [14], and random oversampling. The resulting class distributions after applying the resampling techniques are presented in Table 7. Class weighting was included as a baseline because it does not alter the training-set distribution, whereas the other strategies modify the distribution of the minority classes.

TABLE 7. Class Distribution After Applying Resampling Strategies

Strategy	Lower	Middle	No_income	Upper
SMOTE	262,370	262,370	262,370	262,370
ADASYN	264,054	264,329	262,370	263,780
SMOTE + Tomek Links	256,395	258,178	259,337	261,982
Random Oversampling	262,370	262,370	262,370	262,370

As shown in Table 7, SMOTE and random oversampling produced an exactly balanced training distribution, with 262,370 observations in each class. In contrast, ADASYN produced a slightly different number of synthetic observations for each minority class because it adaptively generates samples according to local data distributions. SMOTE with Tomek Links resulted in a less balanced distribution because Tomek Links removed observations after oversampling to reduce overlapping or ambiguous samples.

Each of these strategies was trained using a baseline Random Forest classifier. The performance comparison is presented in Table 8. SMOTE achieved the highest macro F1-score of 56.94%, followed by SMOTE with Tomek Links at 56.69% and ADASYN at 55.88%. The class-weighted baseline and random oversampling produced lower macro F1-scores of 54.55% and 54.33%, respectively. Therefore, SMOTE provided the best overall balance across the four income classes according to the macro F1-score.

A closer look at Table 8 shows that SMOTE improved the macro F1-score by 2.39 percentage points compared with the class-weighted baseline. It also achieved the highest accuracy and macro precision among the evaluated strategies. Although class weighting and random oversampling produced higher macro recall than SMOTE, their lower macro F1-scores indicate that the increased recall was not accompanied by a sufficient improvement in precision. Similarly, ADASYN improved macro F1 over the baseline but remained below both SMOTE-based approaches.

The class-level results further explain the advantage of SMOTE. As shown in Table 8, SMOTE increased the F1-score of the Middle class from 50.40% under class weighting to 57.80%, while maintaining a high F1-score for the No_income class at 88.40%. However, the Upper class remained difficult to classify, with an F1-score of only 16.90%. This indicates that resampling improved the overall balance of the classification task but could not fully resolve the difficulty associated with the highly underrepresented Upper class. The boundary sensitivity of the Upper-income class is further discussed in Subsection IV-C.

Overall, SMOTE was selected as the resampling strategy for the subsequent experiments because it achieved the highest macro F1-score among all evaluated approaches. This choice prioritizes balanced performance across the four income classes rather than overall accuracy, which is particularly important for the imbalanced target distribution considered in this study.

C. CONFUSION MATRICES

This section presents the confusion matrices for all models reported in Table 6. The confusion matrices in Fig. 7 provide further insight into the classification behavior of the evaluated models, particularly with respect to class-specific performance. Overall, the models classify the *No_income* class more effectively, which is consistent with its large representation in the dataset. The *Lower* and *Middle* classes also have substantially higher numbers of correctly classified observations, although some confusion remains between these income categories. In contrast, the *Upper* class is consistently the most difficult to identify. This result is consistent with the target distribution shown in Fig. 2, where the *Upper* class represents only 1.1% of the observations. Consequently, the class has relatively few true-positive cases, while the number of

TABLE 8. Performance Comparison of Resampling Strategies

Strategy	Accuracy (%)	Macro Precision (%)	Macro Recall (%)	Macro F1 (%)
Class Weighting	68.50	56.00	65.50	54.55
SMOTE	**71.10**	**57.00**	62.60	**56.94**
SMOTE + Tomek Links	70.80	56.80	62.50	56.69
ADASYN	69.60	56.10	63.30	55.88
Random Oversampling	68.30	56.00	65.50	54.33

false negatives remains high across the models, indicating that many upper-income observations are assigned to other income categories. The false-positive counts for the *Upper* class also vary considerably across models, with Random Forest and Logistic Regression producing particularly high values. These findings demonstrate the effect of severe class imbalance on minority-class recognition and reinforce the importance of using macro-averaged metrics to evaluate overall model performance.

D. ROC-AUC ANALYSIS

Among the evaluated models, the Soft Voting Ensemble achieved the highest ROC-AUC of 90.87%, as reported in Table 6. Therefore, its ROC curves are presented in Fig. 8 to further examine its class-wise discriminative performance. The model demonstrates strong discrimination across all four income classes, with the *No_income* class achieving the highest class-wise ROC-AUC of 0.97. The *Upper* class also achieves a relatively high ROC-AUC of 0.91, followed by the *Middle* and *Lower* classes with ROC-AUC values of 0.89 and 0.86, respectively. The macro-average ROC curve has an AUC of 0.91, indicating strong overall discrimination when each income class is given equal importance. These results suggest that the Soft Voting Ensemble can effectively distinguish between the income classes across a range of classification thresholds.

E. PRECISION–RECALL ANALYSIS

Among the evaluated models, CatBoosting achieved the highest macro-averaged F1-score of 59.90%, as reported in Table 6. Therefore, its precision–recall curves are presented in Fig. 9 to further examine the model's class-specific performance. The *No_income* class achieves the highest area under the precision–recall curve (0.973), indicating strong performance in maintaining high precision across different recall levels. The *Lower* and *Middle* classes achieve areas of 0.694 and 0.673, respectively, showing comparatively moderate precision–recall performance. In contrast, the *Upper* class has a substantially lower area of 0.147, indicating considerable difficulty in maintaining precision as recall increases for this minority class. This result further highlights the impact of the severe class imbalance on minority-class recognition and demonstrates why macro-averaged evaluation is important for assessing performance across all income classes.

F. FEATURE IMPORTANCE ANALYSIS

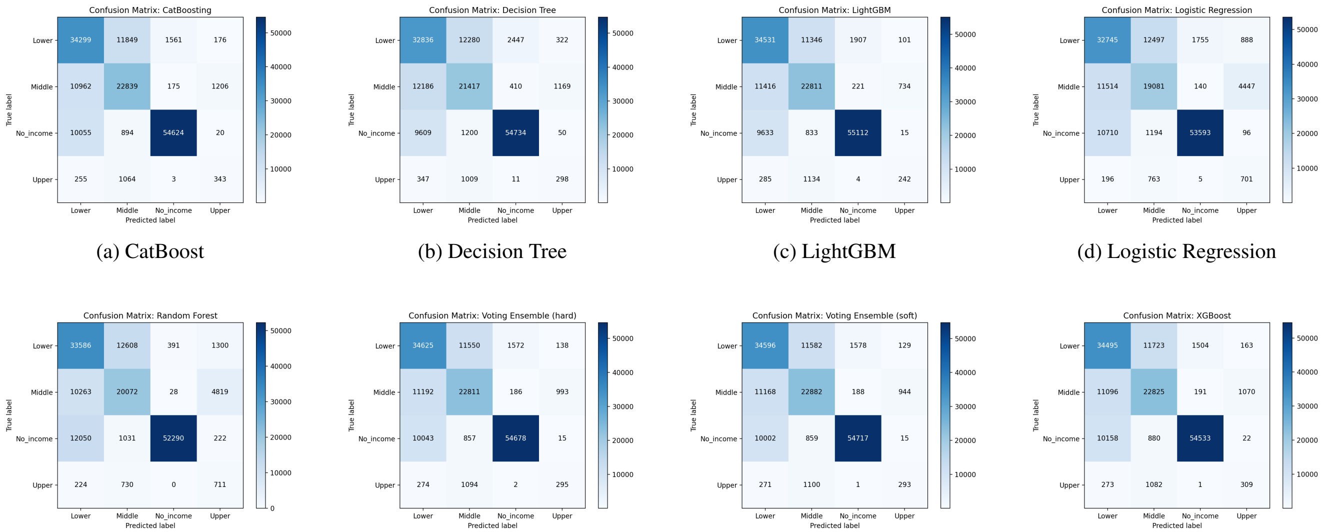
Since CatBoost achieved the highest macro F1-score among the individual models, its feature importance was examined to identify the variables that contributed most to the income-class predictions. Fig. 10 presents the top 15 features ranked by their importance scores.

As shown in Fig. 10, *digital_index* is the most important feature, with an importance score of 14.5. Its leading position indicates that digital access and engagement make the strongest contribution to the model's income-class predictions among all examined variables. This finding suggests that digital connectivity and participation are closely associated with a person's income position and may play an important role in shaping the socioeconomic opportunities reflected in the data. The prominence of *digital_index* also highlights the importance of considering digital inclusion when analyzing income inequality and socioeconomic status.

weekly_workhours is the second most important feature, with a score of 12.8, indicating that work intensity is also strongly associated with the predicted income class. The model further assigns high importance to *workhours_missing* and *age*, with scores of 9.9 and 9.8, respectively. The importance of *workhours_missing* is particularly notable because it indicates that the absence of work-hour information itself carries useful information for distinguishing income classes. This finding is consistent with the treatment of missing work-hour values as informative rather than purely as missing data.

The model also identifies *social_participation_index*, regional indicators, and *gender_male* as moderately important features. In particular, *region_Java* and *region_Sumatra* have importance scores of 7.3 and 6.3, respectively. This suggests that geographic differences are relevant to the model's income-class predictions, potentially reflecting variation in employment opportunities, infrastructure, and economic conditions across regions. Similarly, the importance of *education_years*, with a score of 5.8, indicates that education remains an important socioeconomic characteristic.

The individual indicator *digital_status* has a lower importance score of 0.9, but this does not imply that digital access is unimportant. Rather, its lower contribution may reflect the fact that the broader and more detailed *digital_index* captures digital-related variation more comprehensively. Thus, the feature importance results emphasize that the composite digital measure provides the strongest representation of the relationship between digital engagement and income-class predictions.



(a) CatBoost (b) Decision Tree (c) LightGBM (d) Logistic Regression

(e) Random Forest (f) Hard Voting Ensemble (g) Soft Voting Ensemble (h) XGBoost

FIGURE 7. Confusion matrices of the evaluated machine learning models.

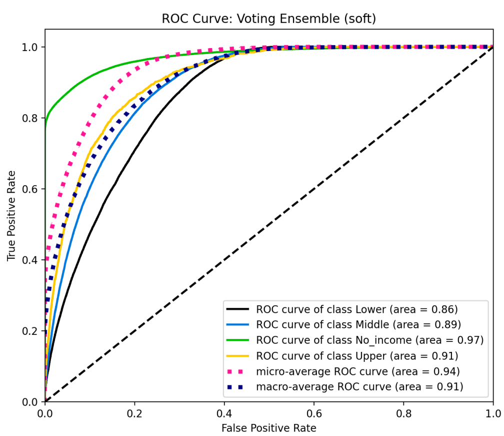


FIGURE 8. ROC curves of the Soft Voting Ensemble for the four income classes.

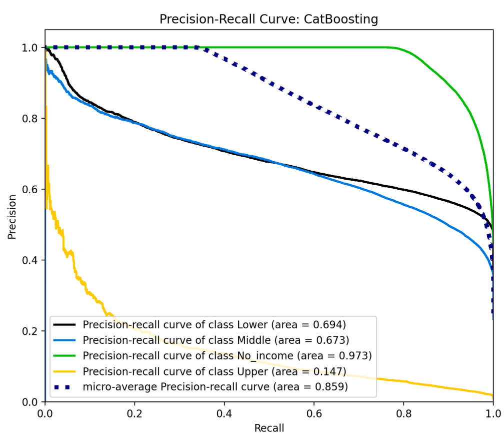


FIGURE 9. Precision–recall curves of the CatBoosting model for the four income classes.

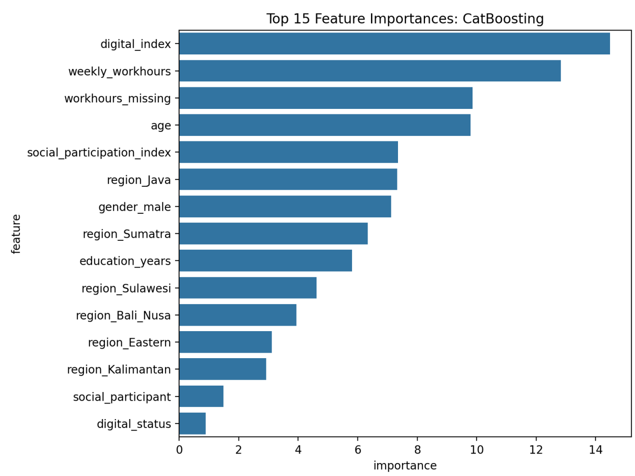


FIGURE 10. Top 15 feature importances of the CatBoost model.

Overall, the feature importance analysis shows that *digital_index* has the highest impact on the CatBoost model's predictions, followed by work-related characteristics, age, social participation, regional factors, and education. These results provide evidence that digital access and engagement are central socioeconomic factors associated with income position in the dataset, while also reinforcing the importance of employment conditions, demographic characteristics, and geographic context. However, these importance scores describe the model's predictive patterns and should not be interpreted as proof that any individual feature directly causes a person's income.

G. EXPLAINABLE AI ANALYSIS

To interpret the model beyond predictive performance, SHAP (SHapley Additive exPlanations) was applied to the best-performing individual model, CatBoost, which achieved a Macro F1-score of 0.599. Furthermore, SHAP values were calculated for 500 test observations across 25 features and four income classes to enable the analysis of both feature-level contributions and feature interactions. The SHAP interaction analysis identified weekly work hours, digital index, education years, and age as particularly influential variables. Their interaction patterns show that the model considers these socioeconomic factors jointly rather than independently. For example, the effect of weekly work hours varies with age and digital engagement, while digital access also interacts with education level as shown in Fig. 11. Additionally, the digital-index dependence plot shows a strong non-linear pattern, where low digital engagement produces negative SHAP contributions and higher digital-index levels increasingly produce positive contributions for the examined class. Besides, the color distribution suggests that higher digital engagement is generally associated with more years of education, which reinforces the relationship between digital access and educational attainment as depicted in Fig. 12.

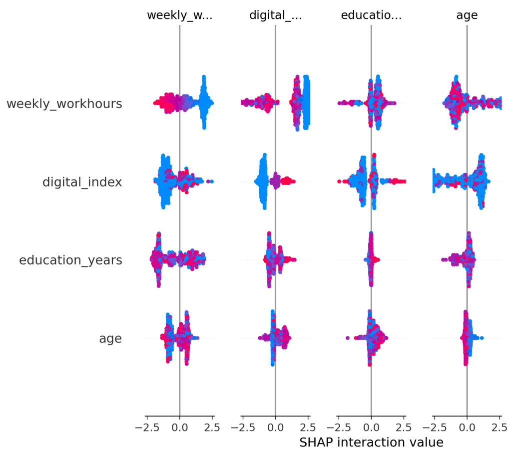


FIGURE 11. SHAP interaction analysis showing the joint effects of weekly work hours, digital index, education years, and age on income-class predictions.

Overall, the XAI analysis indicates that income classification is shaped by the combined effects of employment, digital engagement, education, and age, providing greater transparency into the model's decision-making process.

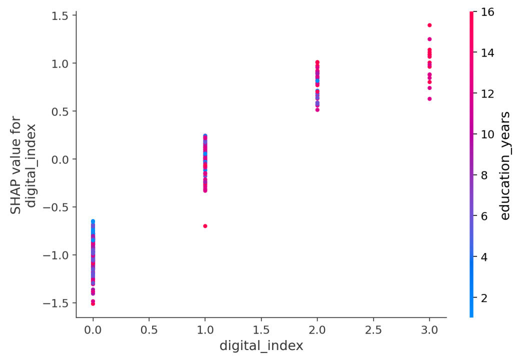


FIGURE 12. SHAP dependence plot of the digital index, showing its non-linear contribution to the examined income class and its association with years of education.

IV. FINDINGS

A. SUBGROUP ANALYSIS

To assess the robustness and potential heterogeneity of classifier performance, subgroup analyses were conducted according to education level, agricultural occupation, disability status, gender, and geographic region. Education level was dichotomized at the sample median. On the other hand, agricultural occupation, disability status, and gender were categorized based on their respective binary indicators. The geographic region was derived from the corresponding one-hot encoded regional variables. To investigate whether the contribution of the digital index to the prediction of the upper-income class varied across population subgroups, SHAP values corresponding to the upper-income class were extracted for the digital index. Mean SHAP values and mean absolute SHAP values were calculated within subgroups defined by education level, agricultural occupation, disability status, gender, and geographic region. Mean SHAP values were used to assess the direction and magnitude of the digital index contribution, while mean absolute SHAP values were used to assess the overall magnitude of its contribution irrespective of direction.

The magnitude of the negative contribution varied across subgroups. Participants with lower education had a more negative mean SHAP value than those with higher education (-0.760 vs. -0.525), with slightly higher mean absolute SHAP values (0.830 vs. 0.794). Agricultural workers also showed a stronger negative contribution than non-agricultural workers (mean SHAP: -0.905 vs. -0.617). However, this finding should be interpreted with caution because only 7 participants were classified as agricultural workers. The contribution was relatively similar between participants with and without disabilities (mean SHAP: -0.721 vs. -0.614).

Differences were also observed by gender. The digital index showed a more negative contribution among females than males (mean SHAP: -0.681 vs. -0.560), accompanied by a higher mean absolute SHAP value (0.840 vs. 0.777). Regional variation was evident, with the strongest negative contribution observed in the Eastern region (mean SHAP = -0.813; mean absolute SHAP = 0.885) and the weakest in Sumatra (mean SHAP = -0.518; mean absolute SHAP = 0.741). Overall, these findings suggest potential heterogeneity in the contribution of the digital index across population subgroups, particularly by education, gender, and geographic region.

Fig. 13 illustrates the subgroup differences in digital-index SHAP values. A statistically significant difference was observed between the lower- and higher-education groups based on the Mann–Whitney U test ($P = 0.0237$). The mean SHAP value was more negative among participants with lower education (-0.760) than among those with higher education (-0.525), which suggests a greater negative contribution of the digital index to Upper-income classification in the lower-education group. No statistically significant differences were observed according to agricultural occupation ($P = 0.0610$), disability status ($P = 0.5128$), or gender ($P = 0.1915$). Although agricultural workers showed a more negative mean SHAP value than non-agricultural workers (-0.905 vs. -0.617), this comparison should be interpreted cautiously given the small number of agricultural workers ($n = 7$).

A statistically significant difference in the distribution of digital-index SHAP values was observed across geographic regions (Kruskal–Wallis $P = 0.0009$). The strongest negative mean SHAP value was observed in the Eastern region (-0.813), whereas the weakest was observed in Sumatra (-0.518). These findings indicate that the contribution of the digital index to Upper-income classification varies across geographic regions and education levels, while no statistically significant subgroup differences were detected for agricultural occupation, disability status, or gender.

B. INTERACTION ANALYSIS

To examine whether the association between digital access and income differs across population groups, interaction effects were analyzed after the subgroup-level SHAP analysis. The SHAP results suggested differences in digital-related feature contributions across education and agricultural-worker groups. However, these differences alone do not establish statistical moderation. Therefore, interaction terms were introduced to formally test whether education level and agricultural-worker status modify the association between digital access and income class.

A multinomial logistic regression (MNLogit) model was used because the target variable contains four income classes: No Income, Lower, Middle, and Upper. No Income was used as the reference category. Continuous predictors, including digital index, education years, age, weekly working hours, and social participation index, were standardized before estimation to improve numerical stability. Binary predictors, including gender, disability status, and agricultural-worker status, were retained in their original form. The models were estimated using maximum likelihood with the L-BFGS optimizer.

The first model examined the interaction between *digital_index* and *education_years*, while controlling for age, weekly working hours, gender, disability status, agricultural-worker status, and social participation. The model converged successfully after 180 iterations using 751,622 observations and achieved a McFadden pseudo-R^2 of 0.4318. As shown in Table 9, the interaction coefficient was positive and statistically significant for Lower, Middle, and Upper income relative to No Income. The coefficients were $\beta = 0.1977$ ($p < 0.001$), $\beta = 0.2379$ ($p < 0.001$), and $\beta = 0.2586$ ($p < 0.001$), respectively. Thus, the positive interaction indicates that the association between digital access and higher income classes becomes stronger as education years increase.

The predicted probabilities provide a clearer view of this interaction. At a Digital Index of 0, the predicted probability of Middle income was 4.39% for individuals with 6 years of education and 6.13% for those with 12 years. At a Digital Index of 3, these probabilities increased to 32.24% and 47.96%, respectively. Hence, the predicted Middle-income probability increased by 27.85 percentage points for the 6-year group and by 41.83 percentage points for the 12-year group. This widening difference can be seen in Fig. 14. A similar pattern was observed for Upper income, where the predicted probability increased from 0.02% to 0.66% for the 6-year group and from 0.05% to 2.19% for the 12-year group, as shown in Fig. 15.

These results suggest that education may strengthen the relationship between digital access and income position. At lower levels of digital access, the predicted income probabilities of the two education groups are relatively close. As digital access increases, however, the higher-education group shows a larger shift toward Middle and Upper income classes. Therefore, digital access alone may not provide the same economic advantage across education levels. Higher education may increase the ability to use digital access in ways that are associated with better economic outcomes.

The second interaction model examined whether the association between digital access and income differs between agricultural and non-agricultural workers. The interaction between *digital_index* and agricultural-worker status was negative and statistically significant for Lower versus No Income ($\beta = -2.4337$, $p = 0.0038$) and Middle versus No Income ($\beta = -2.2865$, $p = 0.0066$). However, the interaction was not statistically significant for Upper versus No Income ($\beta = -0.1785$, $p = 0.9325$). The corresponding predicted probabilities are presented in Fig. 16. The negative interaction indicates that the association between digital access and Lower and Middle income classes is weaker among agricultural workers than among non-agricultural workers. However, this difference is not supported for the Upper-income class. This finding suggests that employment context may influence how digital access is associated with economic position. In particular, greater digital access may translate into stronger income-related benefits outside agricultural employment than within it.

Overall, the interaction analysis provides additional evidence for the subgroup differences identified by the SHAP analysis. Education strengthens the association between digital access and income, whereas agricultural employment weakens this association for the Lower and Middle income classes. These findings suggest that digital inequality may involve more than differences in access alone. Individuals may also differ in their ability to convert digital access into economic opportunities. Consequently, digital inclusion programs may benefit from considering education and employment context rather than treating digital access as an isolated factor. However, these findings are associational and should not be interpreted as evidence of causal effects because the data are observational.

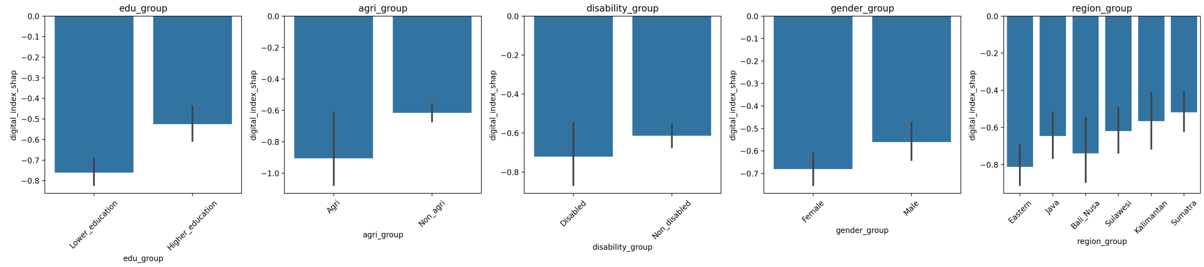


FIGURE 13. Subgroup variation in digital-index SHAP values for Upper-income classification across education, agricultural occupation, disability status, gender, and geographic region.

TABLE 9. Interaction Effects by Income Comparison

Interaction	Income Comparison	Coefficient	*p*-value
digital_index × education_years	Lower vs No_income	0.198	1.18×10^{-165}
digital_index × education_years	Middle vs No_income	0.238	4.99×10^{-210}
digital_index × education_years	Upper vs No_income	0.259	2.49×10^{-77}
digital_index × agri_worker	Lower vs No_income	−2.434	0.004
digital_index × agri_worker	Middle vs No_income	−2.286	0.007
digital_index × agri_worker	Upper vs No_income	−0.178	0.932

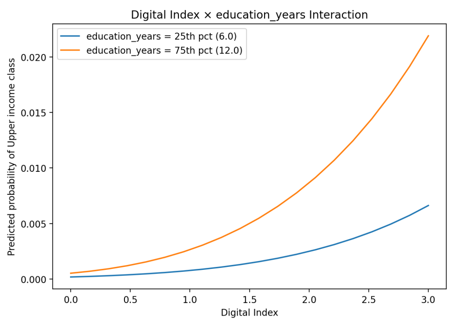


FIGURE 14. Interaction effect of digital index and years of education on predicted upper-class probabilities

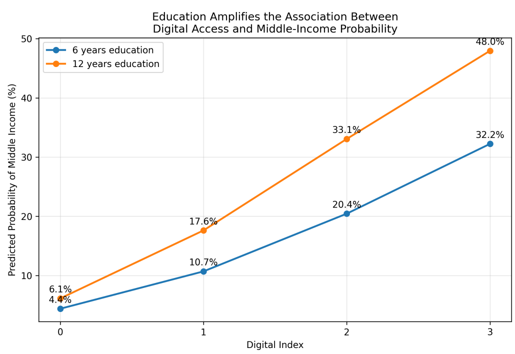


FIGURE 15. Predicted probability of middle-income status by years of education and digital index

C. UPPER-INCOME BOUNDARY SENSITIVITY ANALYSIS

As the upper-income class was defined using a fixed monetary threshold, a sensitivity analysis assessed whether individuals near the boundary differed systematically from those more than 15% above it. Upper-class earners within 15% of the

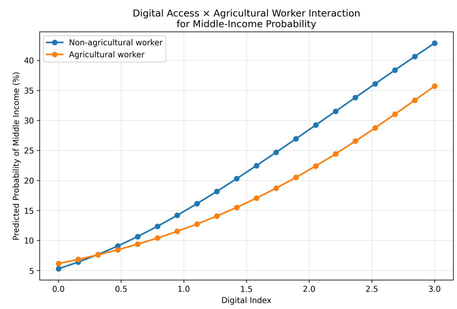


FIGURE 16. Predicted probability of middle-income status by agricultural worker status and digital index

threshold (n = 3,800 of 8,326; 45.6%) were compared using Mann–Whitney U tests [38], with a chi-square test used for regional distribution [39]. Cohen's d was used to quantify the magnitude of differences between the two groups [40]. The significant results are summarized in Table 10.

Age showed the largest effect (Cohen's $d = -0.134$, $p < 0.001$), while *digital_index* and *digital_status* showed statistically significant but very small differences ($|d| = 0.047$–0.063, $p < 0.05$). Regional composition also differed significantly ($\chi^2 = 37.97$, $p < 0.001$). All remaining features showed very small effects and no statistically significant differences ($p > 0.05$).

D. DISABILITY INSIGHT

To examine whether disability patterns differ across income groups, we calculated the percentage of individuals with six types of disability: vision, mental, walking, hearing, communication, and hand/finger disability within each income class. Fig. 17 shows clear differences in disability patterns across income groups. Mental and walking disabilities are particularly

TABLE 10. Significant Results of the Upper-Income Boundary Sensitivity Analysis

Feature/Test	Mean Difference	Effect Size	p-value
Age	-1.4468	$d = -0.1336$	< 0.001
digital_index	-0.0611	$d = -0.0627$	0.00476
digital_status	-0.0300	$d = -0.0465$	0.03704
Region	—	$\chi^2 = 37.97$	< 0.001

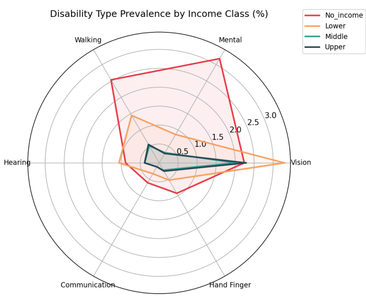


FIGURE 17. Radar plot showcasing the disability prevalence across income groups, highlighting differences in mental, walking, hand/finger, and vision disabilities.

concentrated among individuals in the no-income group. The concentration of mental disability among people with no income may indicate that mental health-related limitations can restrict individuals' ability to participate in the workforce and earn an income. Similarly, the relatively higher prevalence of walking disability in this group suggests that mobility limitations may create barriers to employment and economic participation. Hand/finger disability also appears more prominent among individuals with no income which highlights the relationship between functional limitations and economic disadvantage. In contrast, vision disability appears to be more prominent among the lower-income group rather than being concentrated exclusively among those with no income. For most other disability types, the prevalence generally becomes lower as income increases. Overall, the figure suggests that disability and economic status are closely related, but the relationship varies across disability types. These differences indicate that treating disability simply as a single binary characteristic may obscure important socioeconomic patterns and the distinct ways in which different forms of disability may be associated with income and participation in the workforce.

E. CONCLUSION

This study developed an integrated machine-learning and explainable artificial intelligence framework to investigate income-class differences using demographic, educational, employment, regional, and digital-access characteristics from the August 2023 Indonesian National Labor Force Survey (SAKERNAS). The primary objective was to determine whether the digital index represents the most influential factor in distinguishing individuals across income classes, while also identifying the broader socioeconomic characteristics that contribute to income-class assignment. A comprehensive set of machine-learning models which includes Logistic Regression, Random Forest, XGBoost, CatBoost, and LightGBM, was systematically compared and optimized. SMOTE and ADASYN were further evaluated to address class imbalance, while hard-voting and soft-voting ensembles were examined to assess whether combining strong classifiers could improve predictive performance. This modeling framework demonstrates that multiclass income classification can be effectively approached using a combination of conventional statistical learning, tree-based machine learning, resampling techniques, and ensemble methods. The explainable AI analysis provided important insights beyond predictive accuracy. SHAP analysis of the best-performing CatBoost model identified weekly work hours, digital index, years of education, and age as among the most influential predictors of income-class assignment. The digital index therefore emerged as an important component of income prediction. However, its influence should be understood within a broader socioeconomic structure rather than as an isolated determinant. The findings indicate that employment conditions, human capital, demographic characteristics, and digital access jointly contribute to differences in income classification. The SHAP dependence analysis further revealed a non-linear relationship between digital access and model predictions. Higher digital-index values are generally associated with increasingly positive contributions to the examined income class. Importantly, the interaction analysis showed that the relationship between digital access and income varies across population groups. The positive Digital Index and Education interactions indicate that higher educational attainment strengthens the association between digital access and higher income classes. Conversely, the negative Digital Index and Agricultural Worker interactions for Lower and Middle income classifications suggest that agricultural workers may face greater constraints in converting digital access into income-related advantages. The subgroup analysis provides further evidence of heterogeneous digital effects. Although negative SHAP contributions were more pronounced among individuals with lower education, females, agricultural workers, and those from the Eastern region, statistically significant differences were observed for education level (P=0.0237) and geographic region (P=0.0009). Differences by agricultural occupation, disability status, and gender were not statistically significant. These results emphasize that the socioeconomic value of digital access is influenced by the context in which technology is accessed and used. Taken together, the findings provide a nuanced assessment of the central research hypothesis. The digital index has a substantial predictive contribution to income-class assignment, but the evidence does not justify viewing digital access as a universally dominant factor independent of other socioeconomic characteristics. Instead, its contribution is closely connected with education, employment, age, and regional context. The results therefore support a broader interpretation of the digital divide: inequality is not limited to whether individuals have access to digital technologies, but also concerns their ability to use those technologies to obtain skills, employment opportunities, information, and economic returns. These findings have important implications for policy and development practice. Expanding digital connectivity remains important, but infrastructure alone may not be sufficient to reduce socioeconomic inequality. Digital inclusion strategies should be complemented by education, digital-literacy and technical-skills development, employment opportunities, and targeted socioeconomic support. Particular attention may be required for groups and regions where the ability to translate digital access into economic benefits appears more limited. The proposed analytical framework can assist government agencies and development organizations in identifying such patterns and prioritizing interventions, although the model results should be interpreted as predictive and associational evidence rather than proof of causal effects. Overall, this study demonstrates the value of combining machine learning prediction, class imbalance techniques, ensemble learning, SHAP explainability, and interaction analysis to examine the multidimensional nature of income inequality. The evidence suggests that digital access is one of the important factors of socioeconomic opportunity, but its potential economic access benefits depend substantially on educational, occupational, and regional circumstances of individuals. Hence, a more inclusive digital transformation requires not only expanding access to technology but also strengthening the capabilities and opportunities that enable individuals to convert digital resources into sustainable economic advancement.

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